

Civil Engineering

Trigonometry and Complex Numbers

Revision Booklet

Use these questions for CAT preparation, class discussion and summative revision.

Formula Recall

1. State where the formula below can be used and create one example.

$$\sin \theta = \text{opposite/hypotenuse}$$

2. State where the formula below can be used and create one example.

$$\cos \theta = \text{adjacent/hypotenuse}$$

3. State where the formula below can be used and create one example.

$$\tan \theta = \text{opposite/adjacent}$$

4. State where the formula below can be used and create one example.

$$a/\sin A = b/\sin B$$

5. State where the formula below can be used and create one example.

$$c^2 = a^2 + b^2 - 2ab \cos C$$

6. State where the formula below can be used and create one example.

$$z = a + bi$$

Structured Revision Questions

1. A workplace task involves triangulation. Select the correct mathematical method, solve the problem and write a brief interpretation.
2. A workplace task involves setting out angles. Select the correct mathematical method, solve the problem and write a brief interpretation.
3. A workplace task involves bearings. Select the correct mathematical method, solve the problem and write a brief interpretation.
4. A workplace task involves engineering phasor interpretation. Select the correct mathematical method, solve the problem and write a brief interpretation.
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References and Further Reading

These notes are original learning materials prepared for Mr Mwirigi Mathematics Training Hub. They are academically informed by standard engineering mathematics texts and the local TVET curriculum framework. No textbook pages or worked examples have been copied.

1. Stroud, K. A. and Booth, D. J. Engineering Mathematics, 8th edition. Bloomsbury Academic.
2. Bird, J. Engineering Mathematics. Routledge / Taylor & Francis.
3. Bird, J. Higher Engineering Mathematics. Routledge / Taylor & Francis.
4. Relevant Civil Engineering, Building Technology and Land Survey curriculum and occupational standard documents.